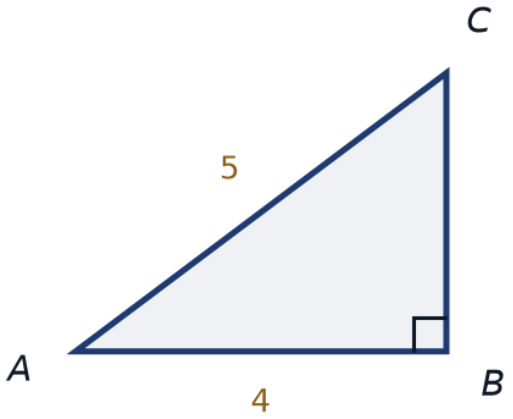
Advanced Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

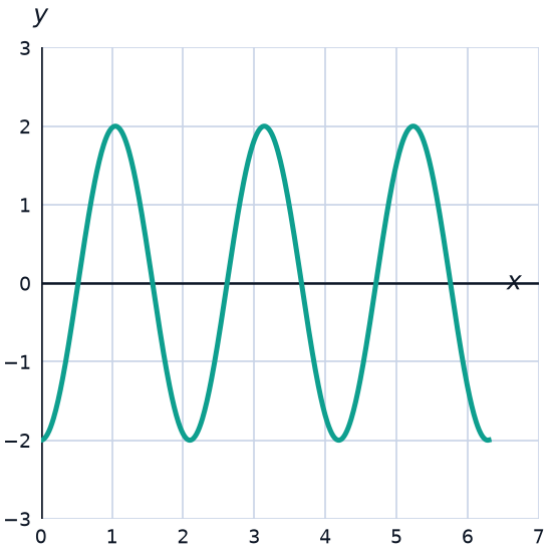
1 [Foundational] • Target: Trigonometric identities: Pythagorean and reciprocal

Given $\cos\theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin\theta$ and $\tan\theta$.



2 [Foundational] • Target: Graphs and transformations of trigonometric functions

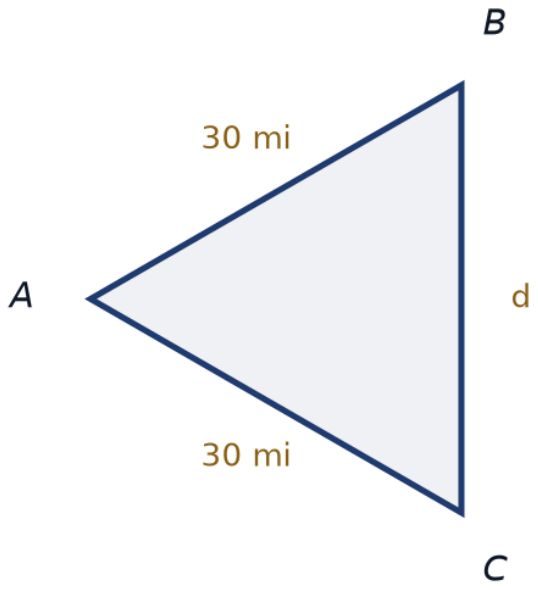
State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.



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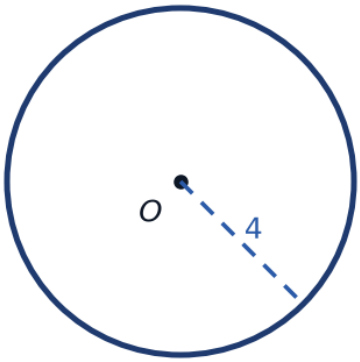
3 [Foundational] • Target: Applications: Law of Sines and Law of Cosines together

Two ships leave a port at the same time, one traveling at 20° and the other at 80° (measured from north), each at 15 mph. Find the distance between the ships after 2 hours, to the nearest mile, using the Law of Cosines.



4 [Foundational] • Target: Polar coordinates

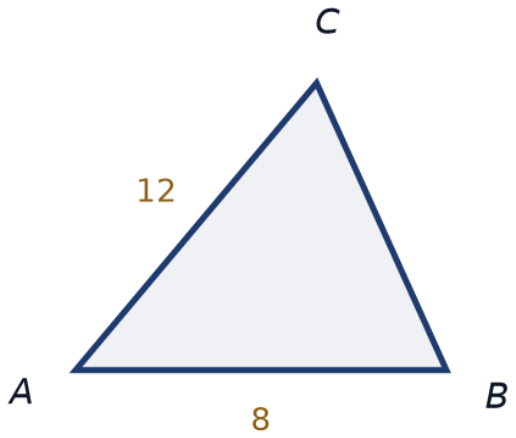
Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.



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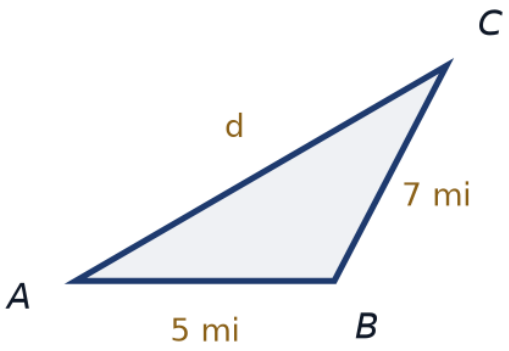
5 [Foundational] • Target: Area of a triangle using trigonometry

Find the area of a triangle with sides 8 and 12, and included angle 50°, using $\text{Area} = \frac{1}{2}ab\sin C$.



6 [Foundational] • Target: A multi-part navigation and triangulation problem

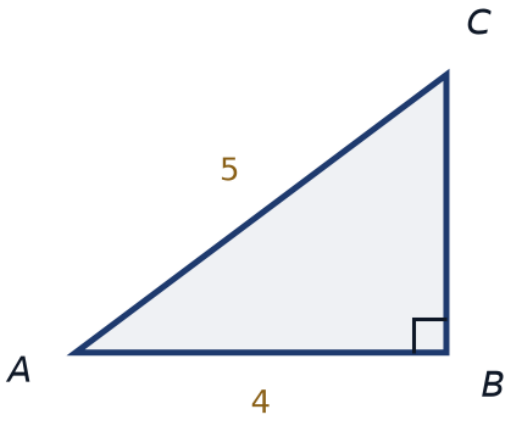
This question has three parts. A hiker walks 5 miles due east, then turns and walks 7 miles at a bearing of 40° north of east. (a) Find the angle between the two legs of the hike (measured as the interior angle at the turning point). (b) Use the Law of Cosines to find the straight-line distance from the start to the end point, to the nearest tenth of a mile. (c) Explain why this distance is less than the total distance walked (12 miles).



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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

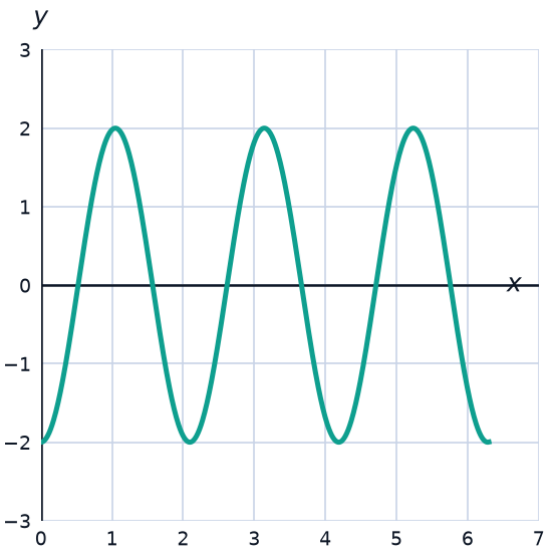
Given $\cos\theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin\theta$ and $\tan\theta$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

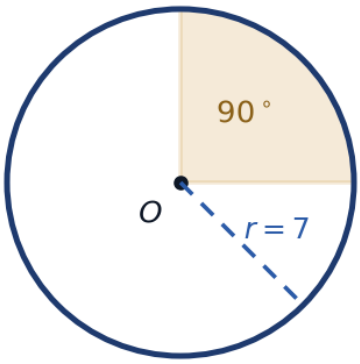
State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.



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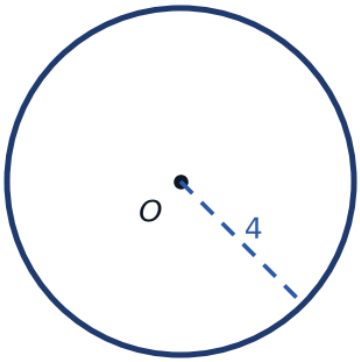
9 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the exact value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

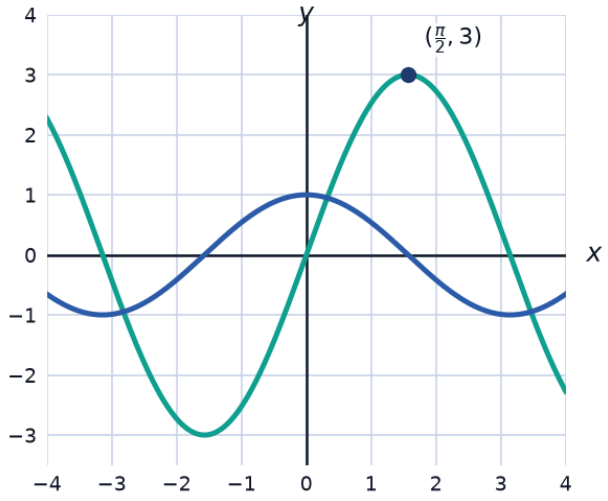
Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.



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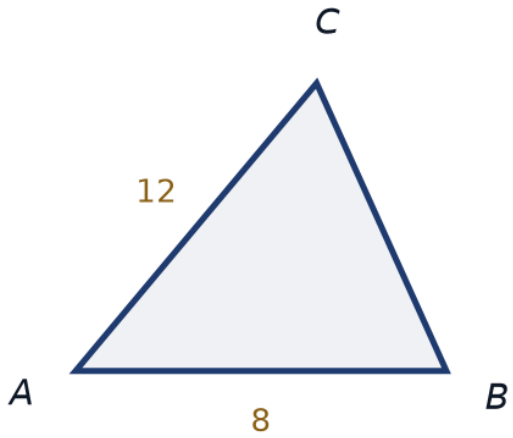
11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

This question has two parts. (a) Use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan\theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan\theta = 3$.



12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the area of a triangle with sides 8 and 12, and included angle 50°, using $\text{Area} = \frac{1}{2}ab\sin C$.



13 [Intermediate] • Target: Sum and difference identities

Use the sum identity to find the exact value of $\sin(75^\circ)$, using $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$.

14 [Intermediate] • Target: Double angle identities

Given $\sin\theta = \frac{3}{5}$ and θ in the first quadrant, find $\sin(2\theta)$ and $\cos(2\theta)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the exact value of $\arcsin\left(\frac{1}{2}\right)$.

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16 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the exact value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$.

17 [Challenging] • Target: Trigonometric equations with multiple solutions • Cross-Lesson: 7.01 Advanced Plane Geometry

Solve $2\cos^2\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

18 [Challenging] • Target: A multi-part trigonometric identity verification problem • Cross-Lesson: 8.01 Statistics, Counting & Probability

This question has two parts. (a) Use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan\theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan\theta = 3$.

19 [Challenging] • Target: Half-angle identities • Cross-Lesson: 1.01 Advanced Number Theory

Use the half-angle identity to find $\cos(15^\circ)$, given $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.

20 [Challenging] • Target: Solving trigonometric equations that factor • Cross-Lesson: 2.01 Advanced Algebra & Systems

Solve $2\sin^2\theta - \sin\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Trigonometric identities: Pythagorean and reciprocal

Given $\sin\theta = \frac{15}{17}$ and θ in the second quadrant, find $\cos\theta$.

a. $-\frac{15}{17}$ **b.** $\frac{15}{17}$

c. $-\frac{8}{17}$ **d.** $\frac{8}{17}$

22 [Foundational] • Target: Sum and difference identities

Given $\cos\theta = -\frac{4}{5}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{4}{5}$ **b.** $\frac{3}{5}$

c. $-\frac{4}{5}$ **d.** $-\frac{3}{5}$

23 [Foundational] • Target: Double angle identities

Given $\cos\theta = -\frac{7}{25}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{24}{25}$ **b.** $-\frac{24}{25}$

c. $\frac{7}{25}$ **d.** $-\frac{7}{25}$

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24 [Foundational] • Target: Graphs and transformations of trigonometric functions

Given $\cos\theta = \frac{12}{37}$ and θ in the first quadrant, find $\sin\theta$.

a. $\frac{12}{37}$

b. $\frac{35}{37}$

c. $-\frac{35}{37}$

d. $-\frac{12}{37}$

25 [Foundational] • Target: Inverse trigonometric functions

Given $\cos\theta = -\frac{8}{17}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{8}{17}$

b. $-\frac{8}{17}$

c. $-\frac{15}{17}$

d. $\frac{15}{17}$

26 [Foundational] • Target: Trigonometric equations with multiple solutions

Given $\cos\theta = -\frac{35}{37}$ and θ in the third quadrant, find $\sin\theta$.

a. $\frac{12}{37}$

b. $-\frac{35}{37}$

c. $-\frac{12}{37}$

d. $\frac{35}{37}$

27 [Foundational] • Target: Applications: Law of Sines and Law of Cosines together

Given $\cos\theta = \frac{12}{13}$ and θ in the first quadrant, find $\sin\theta$.

a. $-\frac{5}{13}$

b. $-\frac{12}{13}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

28 [Foundational] • Target: Polar coordinates

Solve $2\cos\theta = \sqrt{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $225^\circ, 315^\circ$

c. $45^\circ, 315^\circ$

d. $45^\circ, 135^\circ$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A multi-part trigonometric identity verification problem

Solve $\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

a. $225^\circ, 315^\circ$

b. $45^\circ, 135^\circ$

c. 90°

d. 0°

30 [Intermediate] • Target: Half-angle identities

Solve $2\sin\theta = -\sqrt{3}$ for $0^\circ \le \theta < 360^\circ$.

a. $30^\circ, 330^\circ$

b. $240^\circ, 300^\circ$

c. $120^\circ, 240^\circ$

d. $60^\circ, 300^\circ$

31 [Intermediate] • Target: Solving trigonometric equations that factor

Solve $2\sin\theta = -\sqrt{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $120^\circ, 240^\circ$

b. $45^\circ, 135^\circ$

c. 0°

d. $225^\circ, 315^\circ$

32 [Intermediate] • Target: Area of a triangle using trigonometry

Solve $2\sin\theta = \sqrt{3}$ for $0^\circ \le \theta < 360^\circ$.

a. $60^\circ, 120^\circ$

b. 180°

c. 90°

d. $30^\circ, 330^\circ$

33 [Intermediate] • Target: Simplifying trigonometric expressions using identities

Solve $2\cos\theta = -1$ for $0^\circ \le \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. 180°

c. $120^\circ, 240^\circ$

d. $150^\circ, 210^\circ$

34 [Intermediate] • Target: A multi-part navigation and triangulation problem

Given $\sin\theta = \frac{20}{29}$, find $\cos(2\theta)$.

a. $\frac{840}{841}$

b. $-\frac{41}{841}$

c. $\frac{41}{841}$

d. $-\frac{840}{841}$

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35 [Intermediate] • Target: Verifying a trigonometric identity

Given $\sin\theta = \frac{5}{13}$, find $\cos(2\theta)$.

a. $-\frac{120}{169}$

b. $\frac{119}{169}$

c. $\frac{120}{169}$

d. $-\frac{119}{169}$

36 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two sides of a triangle measure 16 and 16, with an included angle of 60°. Find the length of the third side, using the Law of Cosines.

a. 23

b. 16

c. 15

d. 17

37 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two sides of a triangle measure 3 and 8, with an included angle of 60°. Find the length of the third side, using the Law of Cosines.

a. 8

b. 7

c. 6

d. 9

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the area of a triangle with sides 12 and 12, and included angle 150°, using $A = \frac{1}{2}ab\sin C$.

a. 72

b. $-36\sqrt{3}$

c. 37

d. 36

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

Find the area of a triangle with sides 14 and 9, and included angle 150°, using $A = \frac{1}{2}ab\sin C$.

a. $-\frac{63}{2}\sqrt{3}$

b. 63

c. $\frac{63}{2}$

d. $\frac{65}{2}$

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Given $\cos\theta = -\frac{4}{5}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $-\frac{3}{5}$

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Advanced Functions

Given $\cos\theta = -\frac{12}{13}$ and θ in the third quadrant, find $\sin\theta$.

a. $-\frac{12}{13}$

b. $\frac{12}{13}$

c. $-\frac{5}{13}$

d. $\frac{5}{13}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Given $\cos\theta = \frac{24}{25}$ and θ in the first quadrant, find $\sin\theta$.

a. $\frac{7}{25}$

b. $-\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{24}{25}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Given $\sin\theta = \frac{5}{13}$, find $\cos(2\theta)$.

a. $-\frac{120}{169}$

b. $\frac{119}{169}$

c. $\frac{120}{169}$

d. $-\frac{119}{169}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, what is the exact value of $\sin(2\theta)$?

a. $-\frac{24}{25}$

b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{7}{25}$

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Solve for $x \in [0, \pi]$: $\sqrt{3}\tan(2x) = 1$.

a. $\frac{\pi}{12}, \frac{7\pi}{12}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

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